PDF Report

Team Name: QuarkStack

Project Name: QuarkSketch

Team Members:

Niall Concannon (PO) G00433265

Daniel Balcersak (SM) G00435838

Marko Radojcic (Tester) G00436648

Kamil Ciemniejewski (DevOps) G00433270

Module Name: Project Management

Assignment Name: Release Planning

Lecturer: Joseph Corr

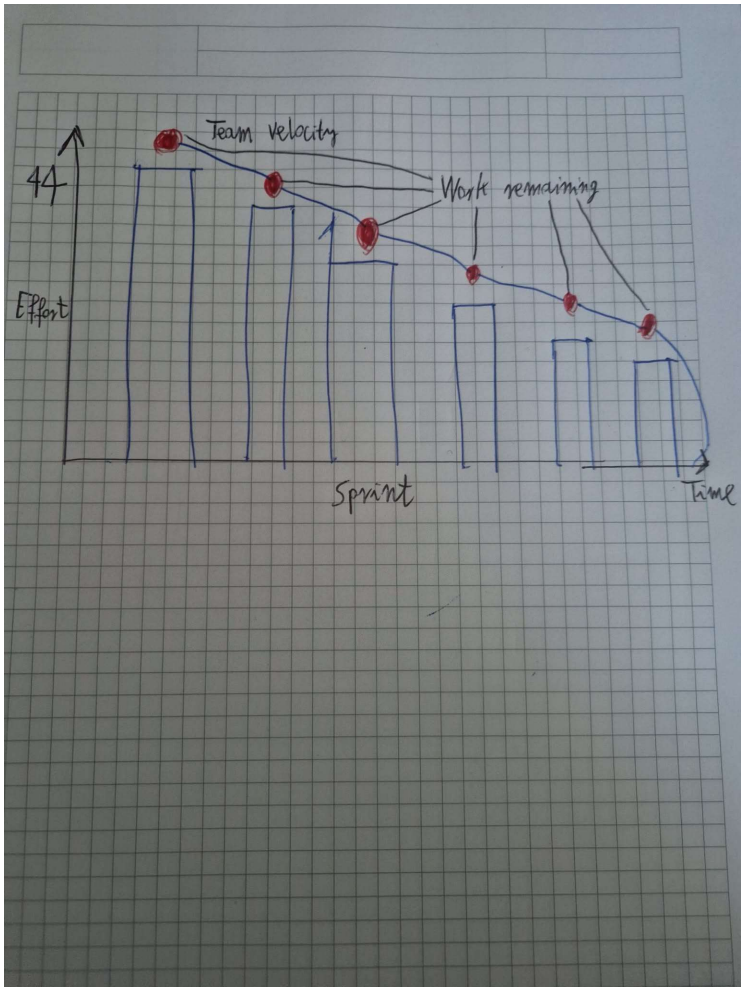
Date: 02/03/2026

Introduction

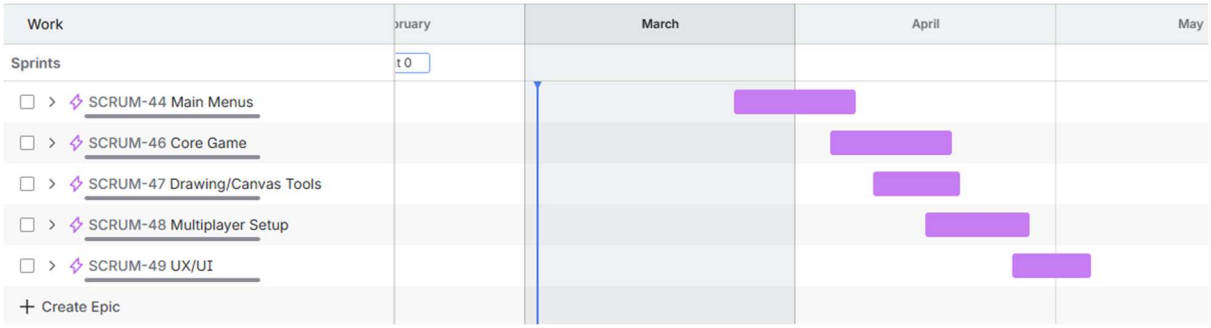
This project is a multiplayer online game where players take turns drawing prompts and guessing each other's drawings. The game features a lobby system, turn-based gameplay, voting for favourite drawings, and point tracking. It is designed to provide an interactive and engaging experience while allowing players to collaborate, compete, and have fun in short rounds.

Release Plan and Burndown Chart

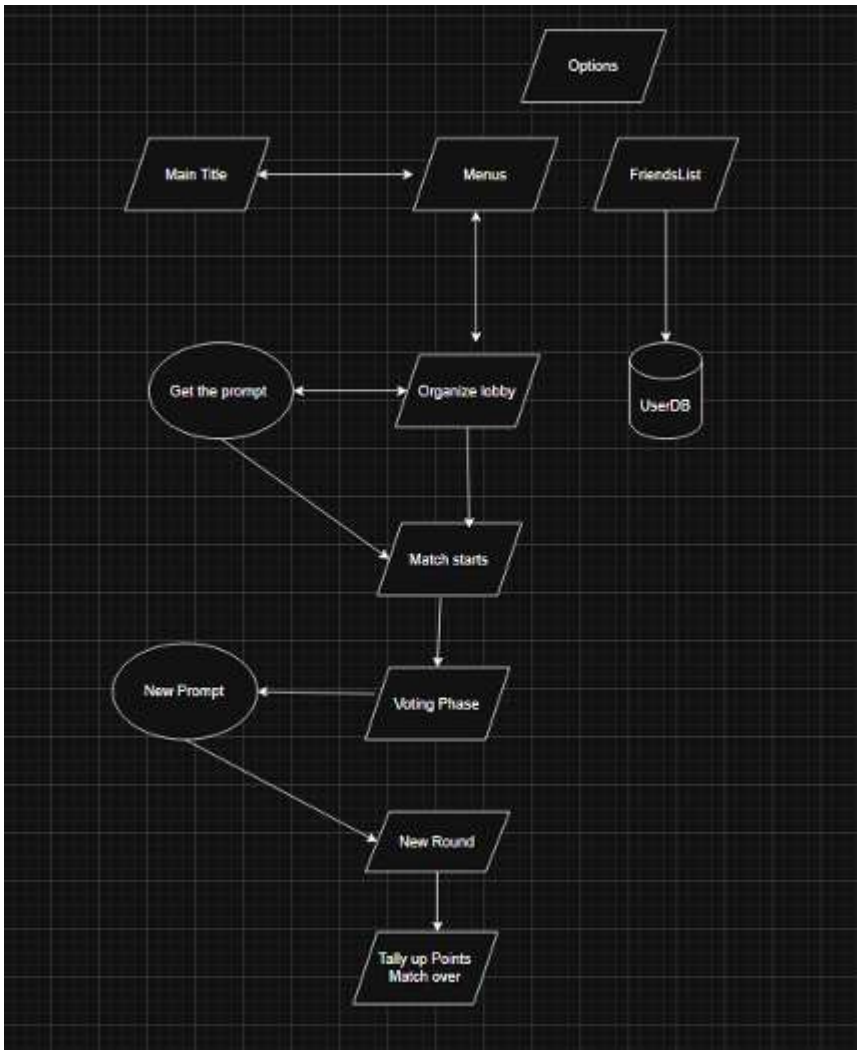
Backlog and Release Plan.xlsx



Product roadmap



System Architecture Overview

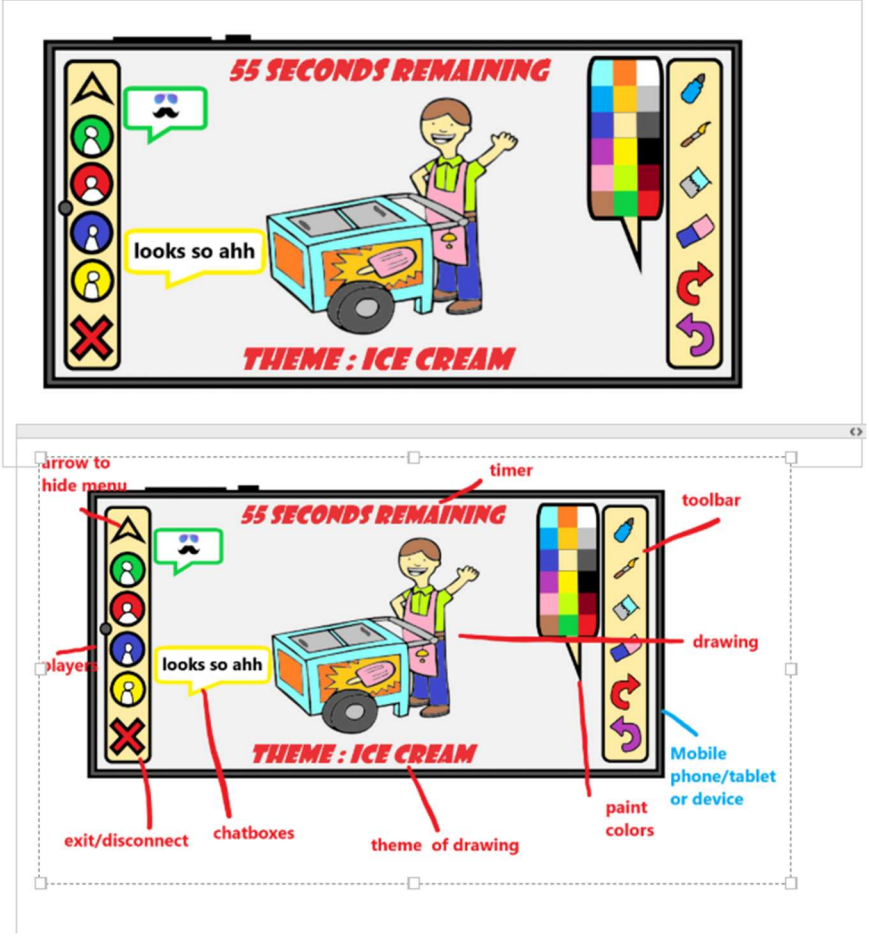


Justification of planning decisions

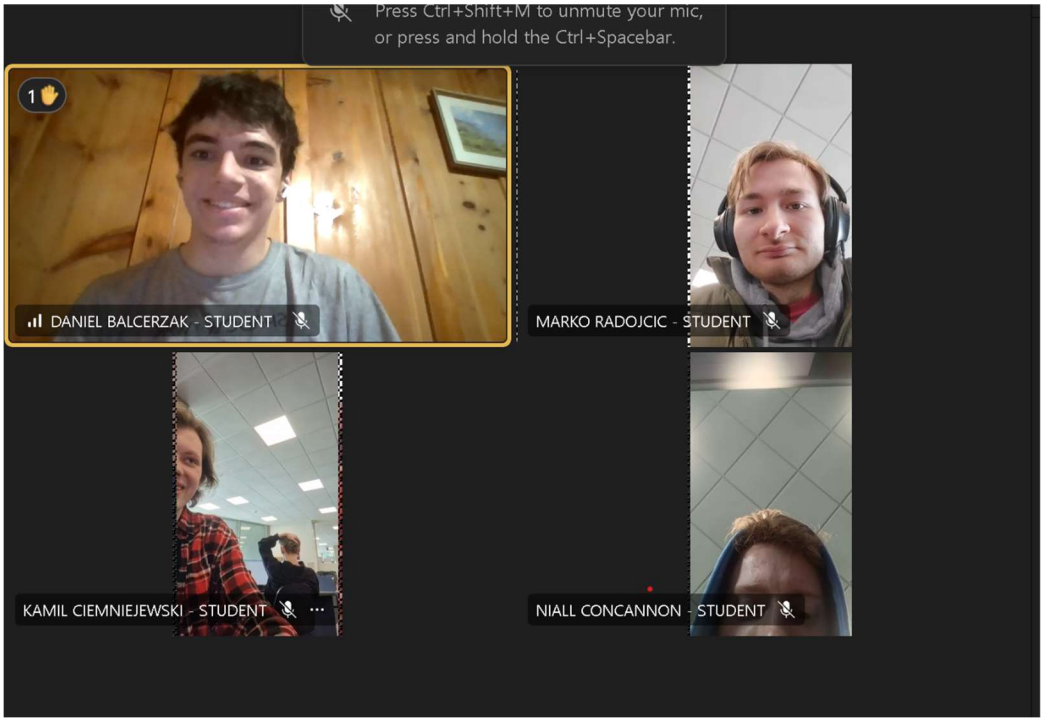
We used a 1-week sprint length to allow fast progress and regular feedback. Short sprints help us test features quickly, identify issues early, and make improvements without delaying the project. Since this is an academic project with limited weekly availability, 1-week sprints help us stay organized and meet deadlines.

Our team consists of 4 members. Each member contributes approximately 3 hours per week, giving us a total of around 12 hours per sprint. Based on our task estimates, we calculated a velocity of approximately 6 story points per 1-week sprint. This ensures the workload remains realistic and achievable within the given timeframe while maintaining steady progress.

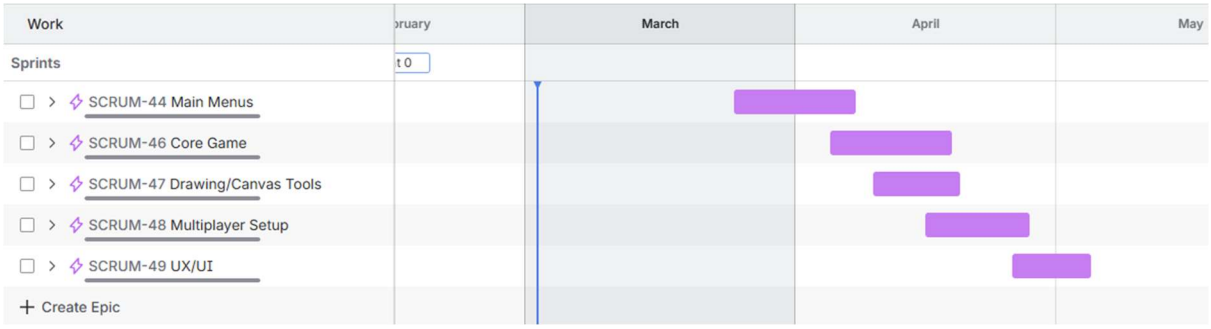
Wireframes



Evidence of meeting



Screenshots of Jira and Gantt charts



Project Risks

1. AI Grading Inaccuracy

Risk Description:
The AI may incorrectly judge drawings or misunderstand prompts. Since AI grading is a core feature of QuarkStack, errors would directly affect gameplay fairness.
Potential Impact:
If the AI scoring feels unfair or inconsistent, players may become frustrated and lose trust in the system. This would reduce engagement and weaken the overall quality of the project.
Impact Level: High
Mitigation Strategy:

- Test the AI with many different prompts and drawings
- Adjust scoring logic based on test results
- Fine-tune thresholds and evaluation criteria
- Perform repeated internal playtesting

Monitoring:

- Log AI scores during testing
- Compare expected vs. actual scores
- Collect structured tester feedback
- Review unusual scoring patterns

2. Multiplayer Connectivity Issues

Risk Description:
As a real-time multiplayer mobile app, QuarkStack may experience lag, synchronization issues, or player disconnections.

Potential Impact:
Network problems could interrupt gameplay, cause lost progress, or make the game feel unreliable. This would significantly harm the user experience.
Impact Level: High
Mitigation Strategy:

- Implement stable client-server communication
- Add reconnection and timeout handling
- Optimize how drawing data is transmitted
- Conduct multiplayer stress testing

Monitoring:

- Track latency and disconnect rates
- Log connection errors
- Test with multiple devices simultaneously
- Monitor response times during gameplay

3. Mobile Performance Problems

Risk Description:
Since the app is written with Angular and typescript and runs on mobile devices, there is a risk of crashes, memory leaks, or performance slowdowns.
Potential Impact:
Poor performance could cause crashes, low frame rates while drawing, or excessive battery usage. This would reduce usability and overall quality.
Impact Level: Medium–High
Mitigation Strategy:

- Regularly check memory management
- Use profiling tools to detect inefficiencies
- Optimize rendering and input handling
- Test on different device types

Monitoring:

- Review crash logs
- Monitor CPU and memory usage
- Perform extended testing sessions
- Conduct performance benchmarking

4. Scope Creep

Risk Description:
Adding too many features (extra game modes, leaderboard expansions, advanced AI improvements) may slow development and reduce focus on core functionality.
Potential Impact:
The team may miss deadlines or deliver unfinished features. Core gameplay quality could suffer.
Impact Level: Medium
Mitigation Strategy:

- Define a clear Minimum Viable Product (MVP)
- Prioritize essential features (drawing, AI grading, leaderboard)
- Avoid adding major features close to deadlines
- Maintain a clear development plan

Monitoring:

- Weekly progress reviews
- Compare progress against timeline
- Track completed vs. planned tasks
- Freeze new features near submission

5. Team Coordination Issues

Risk Description:
As a four-person group project, miscommunication or uneven workload distribution may occur.
Potential Impact:
This could lead to integration problems, delays, or incomplete components at submission time.
Impact Level: Medium
Mitigation Strategy:

- Assign clear roles and responsibilities
- Use version control for code management
- Hold regular meetings
- Keep shared documentation updated

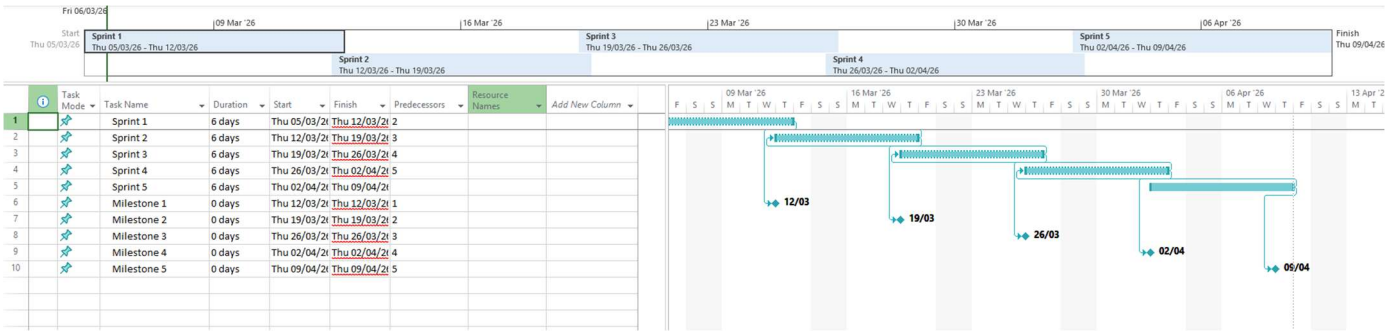
Monitoring:

- Track individual task completion
- Review commits regularly
- Conduct integration testing frequently
- Address delays early

Links

OneNote – https://atlantictu-my.sharepoint.com/personal/g00436648_atu_ie/_layouts/Doc.aspx?sourcedoc={10FA751E-6A86-46C5-835A-779800725507}&end=()&end

Jira – SCRUM Sprint 0 - QuarkStack - Scrum Board - Jira

Github – https://github.com/Niall-Concannon/Project-Management-Group-SA3

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